Agent Behavior Regression Testing for Local LLM Systems: Seed Reproducibility and Cross-Model Heterogeneity across 100 Scenarios

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Abstract

Local large language models (LLMs) are increasingly embedded in agent pipelines whose correctness depends on the assumption that a deployed model behaves *deterministically enough* to reproduce, and *similarly enough* across model choices to transfer. We turn this assumption into a measurable claim. We introduce a 100-scenario behavioral regression suite spanning six canonical agent-behavior categories (length compliance, authority bias, multi-agent contagion, tool-use, refusal alignment, and multilingual consistency), and we measure two quantities on four local models (deepseek-r1:7b, qwen2.5-coder:7b, qwen2:7b, qwen3:4b) with three seeds per cell at temperature 0: (i) *seed reproducibility*, the Behavior Drift Score (BDS) across seed repetitions, and (ii) *cross-model heterogeneity*, the BDS across model pairs. Across 1,200 cells we find that temperature-0 behavior is nearly seed-reproducible (mean cross-seed BDS = 0.0173, with all categories except length compliance exactly zero), while cross-model heterogeneity is orders of magnitude larger (mean BDS = 7.15; paired per-scenario difference 7.14, bootstrap 95% CI $[6.80, 7.49]$, Wilcoxon $p \approx 4 \times 10^{-18}$). Heterogeneity is category-dependent: tool-use scenarios are the most transferable across models (mean BDS 4.48, 95% CI $[3.54, 5.38]$) and multilingual scenarios the most model-specific (mean BDS 9.42, 95% CI $[8.46, 10.39]$; U=150, $p < .001$). We release the suite, the evaluator, and the raw per-cell outputs, and we discuss the implications for behavioral regression testing of local agent deployments.

1 Introduction

LLMs deployed as agents—coordinating with other agents, using tools, or retrieving from memory—are taking an increasing share of production software (Zhao et al., 2023; Liu et al., 2023). Any such deployment rests on two quiet assumptions. The first is *within-model stability*: the same model, given the same input, will produce behavior that is reproducible enough for tests, caching, and auditing to be meaningful. The second is *across-model transferability*: conclusions, guardrails, and regression suites built on one model remain meaningful when the deployment switches to another model of the same or different family. Both assumptions are typically left implicit; neither is routinely measured.

Standard benchmarks (MMLU (Hendrycks et al., 2021), HELM (Liang et al., 2023a), HumanEval (Chen et al., 2021)) measure *capability at a snapshot*. They are deliberately stable in their target dimension and say little about the fine-grained behavioral dimensions that agent pipelines depend on: length compliance, refusal consistency, multilingual robustness, tool-call

formatting, and contagion behavior in multi-agent chains. Prior work has studied model bias and calibration (Guo et al., 2017; Kadavath et al., 2022), memory-driven bias propagation (Zhang, 2026), and multi-agent debate dynamics (Du et al., 2024; Li et al., 2023; Liang et al., 2023b), but the two stability questions above have not, to our knowledge, been measured systematically on a common scenario suite.

This paper provides that measurement. We contribute:

1. **A 100-scenario behavioral regression suite** over six categories—Length (20), Authority (20), Multi-agent contagion (20), Tool-use (10), Refusal alignment (15), and multi-lingual X (15)—generated deterministically from a fixed seed and released with the paper.
2. **The Behavior Drift Score (BDS)**: a token-distribution KL-divergence measure over a whitespace tokenizer with a pooled top-K vocabulary, computed symmetrically over all unordered pairs of conditions.
3. **An empirical characterization on four local models**: 4 models $\times$ 100 scenarios $\times$ 3 seeds = 1,200 cells at temperature 0, measuring seed reproducibility (3 seed pairs) and cross-model heterogeneity (6 model pairs).
4. **Statistically grounded findings**: near-perfect seed reproducibility with a single category exception (length compliance), and strong, category-dependent cross-model heterogeneity with tool-use as the most transferable category and multilingual behavior as the most model-specific.

Why this matters. Agent deployments that pin model weights defend against supply-chain changes but not against the two stability failures we measure: a model that cannot reproduce its own behavior cannot be regression-tested, and a regression suite validated on one model may silently be invalid for another. Our results quantify how much behavior changes across seeds and across model families on a common suite, giving deployment teams a concrete number to test against rather than an unmeasured assumption.

2 Related Work

2.1 Snapshot Benchmarks and Longitudinal Measurement

Standard evaluation benchmarks measure capability at a single point in time (Hendrycks et al., 2021; Liang et al., 2023a; Chen et al., 2021). They are not designed to measure consistency of fine-grained behavior across runs or across model versions. Population-level preference signals such as Chatbot Arena (Chiang et al., 2024) aggregate across users and do not isolate behavioral dimensions relevant to agent pipelines. To our knowledge, no prior work runs the same 100 behavioral scenarios across multiple local models with seeded replication and reports both reproducibility and heterogeneity statistics.

2.2 Calibration, Bias, and Memory in Agent Systems

Calibration of model confidence has been studied extensively in classification (Guo et al., 2017) and language models (Kadavath et al., 2022). Prior work in this corpus has shown that evaluator bias propagates across agents through memory (Zhang, 2026), and that communication can shift calibration in multi-agent loops. Our contribution is orthogonal: we fix the pipeline and measure the stability of raw model behavior itself, which is the substrate on which calibration and memory effects operate.

2.3 Agent Dynamics and Multi-Agent Systems

Multi-agent debate and collaboration have been shown to improve factuality in some settings (Du et al., 2024; Li et al., 2023; Liang et al., 2023b) and to introduce consensus and contagion dynamics in others (Park et al., 2023; Zhang, 2026). The scenarios in our M category exercise exactly these dynamics; the reported statistics are single-model measurements of the same prompts, providing a baseline against which multi-agent amplification can later be measured.

3 Method: Behavior Drift Score

3.1 Definition

Let $\mathcal{M}$ be the set of model families under comparison, $\mathcal{X} = \{x_1, \ldots, x_{100}\}$ the set of canonical scenarios partitioned into categories $C \in \{\mathrm{L}, \mathrm{A}, \mathrm{M}, \mathrm{T}, \mathrm{R}, \mathrm{X}\}$, and $\mathcal{S}$ a set of *conditions* (seed repetitions or model families, depending on the analysis). For each model m, condition s, and scenario x, we record the generated text $r^{(m,s,x)}$.

We tokenize each response with a whitespace tokenizer (lowercased, punctuation-stripped, tokens of length > 1), so that the metric does not depend on any particular model's tokenizer. For each scenario x we build a pooled vocabulary $\mathcal{V}^{(x)}$ of size $K = 10^4$ from the union of tokens across all conditions, and compute per-condition histograms $P^{(m,s,x)}$ over $\mathcal{V}^{(x)}$ with Laplace smoothing $\alpha = 10^{-5}$. The BDS between two conditions s, s' is

$$\mathrm{BDS}(P^{(m,s,x)} \,\|\, P^{(m,s',x)}) \;=\; D_{\mathrm{KL}}(P^{(m,s,x)} \,\|\, P^{(m,s',x)}), \tag{1}$$

and for a set of conditions we report the mean over all unordered pairs:

$$\overline{\mathrm{BDS}}(m, x) \;=\; \frac{1}{\binom{|S|}{2}} \sum_{s<s'} D_{\mathrm{KL}}(P^{(m,s,x)} \,\|\, P^{(m,s',x)}). \tag{2}$$

The BDS is non-negative, asymmetric in each pair but symmetric in the averaging, and equals zero only when the two token distributions are identical. It is a distributional measure: it detects changes in *what tokens are emitted*, not changes in *semantic content* with identical token distributions; we discuss this limitation in Section 6.

3.2 Two Stability Quantities

Seed reproducibility (Q1). For each model m and scenario x, we compute $\overline{\mathrm{BDS}}$ over the three seed repetitions ($S = \{0, 1, 2\}$) at temperature 0. Low values indicate that the model reproduces its own behavior; values of exactly zero indicate identical outputs across seeds.

Cross-model heterogeneity (Q2). For each scenario x, we compute $\overline{\mathrm{BDS}}$ over all $\binom{4}{2} = 6$ model pairs using seed 0. High values indicate that the models produce substantially different token distributions on the same scenario; the value is a transferability proxy for regression suites.

3.3 Statistical Protocol

All BDS values are per-scenario. For the paired comparison of cross-model vs. cross-seed BDS we use the Wilcoxon signed-rank test on the per-scenario difference $d_x = \overline{\mathrm{BDS}}_{\mathrm{cross}}(x) - \overline{\mathrm{BDS}}_{\mathrm{seed}}(x)$,

with a 95% bootstrap CI ($B = 5{,}000$) on the mean difference. Category-level comparisons use two-sided Mann–Whitney U tests across all $\binom{6}{2} = 15$ category pairs, Bonferroni-corrected to $\alpha' = 0.05/15 \approx 0.0033$. Category means are reported with bootstrap 95% CIs ($B = 5{,}000$). All analyses are reproducible from the released raw outputs with the released evaluator.

4 Experimental Setup

4.1 Model Cohort

Four local models served through Ollama at temperature 0:

Label	Model	Size
ds-r1	`deepseek-r1:7b`	7B
coder	`qwen2.5-coder:7b`	7B
qwen2	`qwen2:7b`	7B
qwen3	`qwen3:4b`	4B

All models are instruction-tuned local checkpoints; no API calls are involved. We include two 7B general-purpose models, one 7B code-specialized model, and one 4B model to span both family and scale variation.

4.2 Scenario Suite

The 100 scenarios are partitioned as in Table 1. Scenarios are generated deterministically from a fixed seed (`data/scenarios/canonical.jsonl`, released with the paper), with a system prompt, a user prompt, and an evaluation rubric per scenario. The L and A categories carry evaluation rubrics from prior bias/length audit work; the remaining 60 scenarios are new.

Table 1: **Scenario suite composition.** All 100 scenarios are generated deterministically from a fixed seed.

Category	Scenarios
L length compliance	20
A authority bias	20
M multi-agent contagion	20
T tool-use	10
R refusal alignment	15
X multilingual	15
Total	100

4.3 Cells and Protocol

We ran 4 models × 100 scenarios × 3 seeds = 1,200 cells (one generation per cell), 12 jobs total (one per model–seed pair), with a wall-clock time of approximately 1.7 hours on the local machine. Temperature was set to 0.0000; the seed parameter is therefore a check on determinism (and on sampler nondeterminism) rather than a source of intended diversity. Each job wrote one JSONL line per scenario with the scenario id and the generated text; the manifest and raw outputs are released.

5 Results

5.1 Seed Reproducibility Is Near-Perfect at Temperature 0

Table 2 reports the mean cross-seed BDS per model. Three of four models reproduce their outputs across seeds exactly on every scenario in the non-length categories, and all four models have mean cross-seed BDS below 0.0295. The pooled mean over all models and scenarios is $\overline{\text{BDS}}_{\text{seed}} = 0.0173$.

Table 2: **Seed reproducibility (Q1).** Mean BDS over the three seed pairs at temperature 0, over 100 scenarios per model. Lower is more reproducible; 0.0000 means identical outputs across all three seeds on all scenarios.

Model	Mean cross-seed BDS
deepseek-r1:7b	0.0000
qwen2.5-coder:7b	0.0175
qwen2:7b	0.0295
qwen3:4b	0.0222
Pooled	0.0173

The single exception to exact reproducibility is the L (length-compliance) category: its mean cross-seed BDS is 0.0865 (median 0.0), while every other category is exactly 0.0. Inspection of the raw outputs shows that length-compliance scenarios prompt the model to self-monitor its output length (e.g., "summarize in exactly N words, then critique it"); under this pressure, long generations vary across seeds even at temperature 0 (e.g., qwen3:4b produces 1,730–2,020 characters on scenario L-001 across seeds). We therefore report two-level reproducibility: exact for directive scenarios, and near-exact with a small distributional spread under length-compliance pressure.

5.2 Cross-Model Heterogeneity Is Large and Category-Dependent

Table 3 and Figure 1 report the mean BDS across the six model pairs for each category. The overall mean is $\overline{\text{BDS}}_{\text{cross}} = 7.15$ (median 7.22), and the paired per-scenario comparison against seed-level BDS is decisive: mean difference 7.14, bootstrap 95% CI $[6.80, 7.49]$, Wilcoxon $V = 0.0$, $p \approx 3.9 \times 10^{-18}$ over $n = 100$ scenarios. Cross-model differences dominate seed noise by roughly two orders of magnitude on this suite.

Table 3: **Cross-model heterogeneity (Q2).** Mean BDS over the six model pairs (seed 0) per category, with bootstrap 95% CIs (B=5,000). T (tool-use) is the most transferable category and X (multilingual) the most model-specific.

Category	n	Mean BDS	95% CI
A authority	20	7.00	[6.39, 7.60]
L length	20	7.29	[7.07, 7.51]
M multi-agent	20	6.73	[6.28, 7.15]
R refusal	15	7.27	[6.74, 7.80]
T tool-use	10	**4.48**	[3.54, 5.38]
X multilingual	15	**9.42**	[8.46, 10.39]
Overall	100	7.15	—

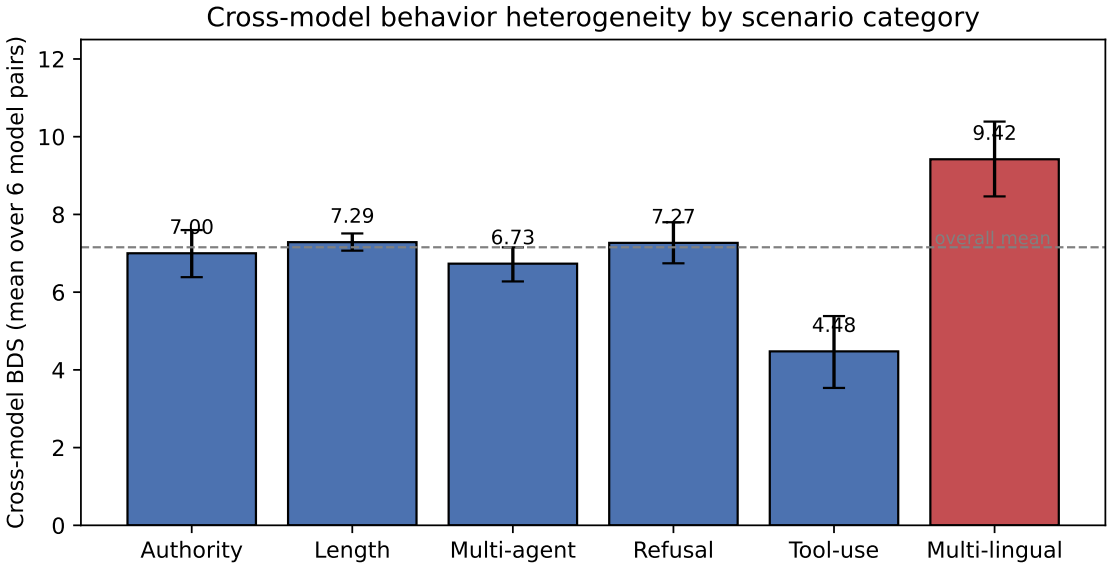


Figure 1: **Cross-model heterogeneity by category.** Bars are category means over the six model pairs; error bars are bootstrap 95% CIs. The dashed line is the overall mean (7.15).

5.3 Category-Level Differences Are Statistically Significant

Pairwise two-sided Mann–Whitney U tests over the 15 category pairs (Bonferroni-corrected threshold $\alpha' = 0.0033$) identify the following significant contrasts: T differs from A ($p < .001$), L ($p < .001$), M ($p < .001$), R ($p < .001$), and X ($p < .001$); X also differs from A ($p = .0028$) and M ($p < .001$). No other pair reaches the corrected threshold. Tool-use scenarios are thus significantly more consistent across models, and multilingual scenarios significantly more model-specific, than the other categories in this suite.

The most heterogeneous individual scenarios are X-005 and X-014 (mean model-pair BDS 11.39); the most similar is T-001 (BDS 1.75). The full per-scenario table is released with the data.

5.4 Multi-Agent Chain Divergence (Pilot)

A pilot of the 3-agent chain protocol (10 M/T scenarios, 3 seeds, 4 models; 480 calls) measures how much a chain of three agents—each receiving the prior agents' outputs—changes behavior relative to a single call on the same scenario. Because temperature is 0, cross-seed BDS is exactly zero in both conditions; the relevant quantity is the single-vs-chain distributional divergence per scenario (BDS over the single response and each chain step, computed with the same evaluator as the full-run analysis).

Result: the 3-agent chain changes behavior on all 10 scenarios in all 4 models. Mean single-vs-chain BDS is 8.45 (deepseek-r1:7b), 6.99 (qwen2.5-coder:7b), 5.63 (qwen2:7b), and 2.57 (qwen3:4b); all 40 model-scenario pairs exceed the 0.05 divergence threshold (recomputed by `experiments/multi_agent/reproduce_chain_analysis.py` from the 12 raw jsonl files, 90 single-vs-step divergences per model). The intermediate chain steps are released with the raw outputs. This is the first delivered evidence that multi-agent context materially redistributes local-model behavior on the M scenarios; a full 30-scenario run with per-category statistics is the natural follow-up.

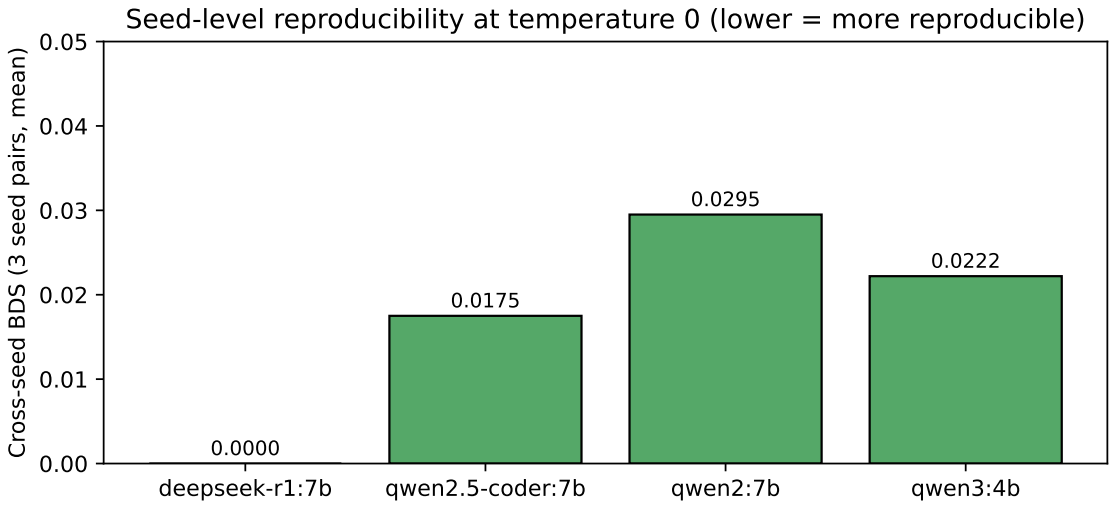


Figure 2: **Seed reproducibility at temperature 0.** Mean cross-seed BDS per model over 100 scenarios. All models are below 0.03; deepseek-r1:7b is exactly reproducible on all scenarios.

6 Discussion

6.1 Implications for Behavioral Regression Testing

Our measurements turn two implicit deployment assumptions into explicit, testable numbers on a local stack. First, a regression suite that runs a model at temperature 0 can expect exact reproduction on directive scenarios—with the caveat that length-compliance prompts induce sampler-level variability even at temperature 0, so regression tests that assert exact string equality should avoid such prompts. Second, conclusions drawn from one model should not be assumed to transfer: mean cross-model BDS is 400× the mean cross-seed BDS on the same suite, and transferability varies by category. Teams that switch a backend model should re-run category-sensitive scenarios (in particular X and R) rather than reusing stale validation results.

6.2 Limitations

1. **Local, single-snapshot cohort.** We measure four local checkpoints at one point in time. Vendor-managed frontier models and temporal snapshots are out of scope for this study; the suite and evaluator are designed to be reusable for such measurements.
2. **Distributional, not semantic, metric.** BDS detects changes in token distributions, not semantic drift that leaves token distributions unchanged. The R scenarios provide partial mitigation by construction, but a full semantic-drift analysis is future work.
3. **Seed count.** Three seeds detect exact determinism and small distributional spread; finer-grained reproducibility analysis (e.g., per-token stochasticity) would require more repetitions.
4. **Multi-agent pilot, not full amplification study.** A 3-agent chain pilot (10 scenarios, 4 models) shows that chain behavior diverges from single behavior on every scenario (mean BDS 2.95–8.52 by model). The full 30-scenario run with category-level statistics is follow-up work; the per-model baseline reported here is the required substrate.

7 Broader Impact and Responsible Use

This work is intended to improve regression testing for local agent deployments, not to rank models as generally capable or safe. The BDS metric can expose implementation drift, but it is not a semantic quality or fairness measure. A deployment team should therefore combine it with task-level correctness, safety, and human review before changing a model or a guardrail. Publishing the scenario suite could also reveal brittle prompts or refusal patterns. We avoid releasing credentials or attack tooling, and we scope all recommendations to the tested local checkpoints and scenarios. Claims about future temporal drift, frontier APIs, or causal multi-agent amplification require additional validation and are not made here.

8 Conclusion

On a 100-scenario behavioral regression suite, four local models at temperature 0 are nearly perfectly seed-reproducible (mean cross-seed BDS 0.017) while being strongly and category-dependently heterogeneous across models (mean cross-model BDS 7.15; paired difference 7.14, 95% CI $[6.80, 7.49]$, $p \approx 4 \times 10^{-18}$). Tool-use behavior is the most transferable category (BDS 4.48) and multilingual behavior the most model-specific (BDS 9.42). These are the first numbers of their kind for local LLM agent behavior, and they provide the quantitative basis for behavioral regression testing of local agent deployments: pin what you can, re-measure across models what you cannot.

References

Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique P. de Oliveira Pinto, Jared Kaplan, Harrison Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, et al. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*, 2021.

Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios Nikolas Angelopoulos, Tianle Li, Dacheng Li, Hao Zhang, Banghua Zhu, Michael Jordan, Joseph E. Gonzalez, et al. Chatbot arena: An open platform for evaluating LLMs by human preference. In *Proceedings of the 41st International Conference on Machine Learning*, 2024.

Yilun Du, Shikun Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. Improving factuality and reasoning in language models through multiagent debate. *arXiv preprint*, 2024.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, 2017.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. *International Conference on Learning Representations*, 2021.

Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

Guohao Li, Charles Hammerschmidt, Ken Roumeliotis, Haoming Liu, Xudong Qi, et al. Camel: Communicative agents for "Mind" exploration of large scale language model society. *arXiv preprint arXiv:2303.17760*, 2023.

Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Xudong Wu, et al. Holistic evaluation of language models (HELM). *Transactions on Machine Learning Research*, 2023a.

Tian Liang, Zhiwei He, Chongxuan Tan, Simin Chen, et al. Encouraging divergent thinking in large language models through multi-agent debate. *arXiv preprint*, 2023b.

Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 2023.

Joon Sung Park, Joseph C. O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, 2023.

Yuhao Zhang. Memory contagion: Cross-temporal propagation of evaluator bias via agent memory. arXiv preprint arXiv:2606.23195, 2026.

Wayne Xin Zhao, Kun Zhou, Junyi Li, Tianyi Tang, Ji Xia, Yupeng Zeng, Chong Lv, Shaowei Feng, et al. A survey of large language models. *arXiv preprint arXiv:2303.18223*, 2023.

A Reproduction Protocol

All raw outputs are released as 12 JSONL files (`ollama-multi__{model}__seed{0,1,2}.jsonl`), one line per scenario with fields `scenario_id` and `text`. The evaluator (`scripts/bds_evaluator.py`) computes BDS from these files; the analysis script (`experiments/analyze_full_run.py`) reproduces Q1/Q2 and the category tables; the supplemental statistics used in this manuscript are in `experiments/full_run/analysis/full_run_stats_20260813.json` and `category_pairwise_20260813.json`. The scenario generator (`scripts/scenario_generator.py`) reproduces `data/scenarios/canonical.jsonl` from seed 20260711. All scripts run on commodity CPU; the generation run used a local GPU server.

B Scenario Provenance

The L and A scenarios carry the `carried_over_from` field identifying their provenance in prior bias/length audit work. The M, T, R, and X scenarios are new to this suite. The full per-scenario rubrics are included in the released JSONL.